

Anil Sahith Vallepu

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PROFESSIONAL SUMMARY

Software Engineer with strong Math and CS foundations, specializing in AI/ML systems and production-grade deployment. Proven expertise in developing and optimizing deep learning models, LLM applications, and RAG systems. Skilled in building low-latency inference pipelines, on-device model optimization, and scalable data architectures.

Seeking roles: Software Engineer, AI Engineer, Machine Learning Engineer, Data Scientist, Quant Developer.

SKILLS

- **Languages & Tools:** Python, C++, JavaScript, TypeScript, SQL, GitHub, Postman, Cursor.
- **Web Technologies:** React.js, Next.js, Node.js, Django, Flask, FastAPI, WebSockets, Temporal, Kafka, ZeroMQ.
- **Databases & Infra:** MySQL, PostgreSQL, MongoDB, Redis, Pinecone, MinIO, AWS, GCP, Docker, K8, Modal, Supabase.
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, PySpark, OpenCV, NLP, LLMs, VLMs, RL, RAG.
- **Other Coursework:** Data Structures and Algorithms, System Design, Embedded Systems, AI Voice Agents.

EXPERIENCE

- Matters.AI** Dec 2025 - Current
Software Engineer - AI/Backend *Bangalore, Karnataka*
 - Worked on a production-grade data processing system handling massive datasets (unstructured & structured) using Dask for distributed parallel execution and a hybrid extraction engine for PII/PHI detection, integrating VLM and LLM models for image extraction, entity classification, and post-verification.
 - Designed and deployed a 'Universal Rate Limiter' proxy service (Redis Streams + AsyncIO) on a GKE cluster (Google Kubernetes engine), decoupling upstream ingestion (up to 10k RPS) from throttled downstream Modal AI inference (200 RPS), achieving ~0.01% failure rates and P95/99 latency of ~10-15s under peak batch processing loads via adaptive controllers. Infrastructure includes Redis on AWS ElastiCache with monitoring via New Relic (CubeAPM).
 - Optimized NER and VLM model deployment for improved overall throughput and CPU-GPU utilization through A/B testing, achieving optimal cost efficiency on the Modal platform while maintaining strict SLA guarantees.
 - Noccarc Robotics Pvt Ltd** June 2024 - April 2025
AI/ML Engineer *Pune, Maharashtra*
 - Architected a real-time arrhythmia detection system for embedded patient monitoring devices using deep learning, established an end-to-end data pipeline from sensor integration to frontend visualization with automated alarm generation across 20+ distinct classifications for critical heart abnormalities, reducing clinical response times in ICUs.
 - Implemented a hybrid neural network combining ResNet, CBAM attention mechanism, and GRU layers with separable convolutions for efficient edge deployment, achieving ~300ms latency on a patient monitor(edge-device) while processing ECG data every 5 seconds and consuming under 12% single-core CPU resources.
 - Carelon Global Solutions (Elevance Healthcare)** June 2023 - August 2023
Summer Intern - AI/ML *Bangalore, Karnataka*
 - Automated document processing pipeline using computer vision and deep learning to extract and classify data from 700+ documents per hour with 95% accuracy; deployed an end-to-end OCR solution using PyTesseract for text extraction, OpenCV for image preprocessing, and YOLOv8 object detection for region identification.
 - Accelerated processing performance through multi-threading and multi-processing techniques, reducing document processing time by 90% and decreasing operational costs, delivered a production-ready solution by integrating with existing data workflows for scalable enterprise document management.

PROJECTS

- **CryptoAI - AI Trading Engine:** Built an autonomous crypto futures trading system using a multi-agent architecture, where specialized AI agents collaborate to analyze markets, identify high-probability swing and scalp setups, validate risk parameters, execute trades in real-time via WebSockets, store trade history, and continuously learn from performance, all monitored in a real-time dashboard with live portfolio updates and agent activity feeds.
Tech stack: Python, Next.js, FastAPI, Redis, PostgreSQL, LangGraph, Groq LLM API, Docker, Vercel.
- **SlackBot Content Creation AI Assistant:** Launched an AI-powered Slack bot streamlining content research, keyword analysis, and competitive analysis workflows; incorporated multiple APIs for web scraping, search engine results, and email notifications via SendGrid, reducing content planning time from 6+ hours to 15 minutes for marketing teams.
Tech stack: Python, Supabase, Upstash, SendGrid, Brave Search API/SerpAPI, Scikit-learn, Groq, Docker, Render.

EDUCATION

- National Institute of Technology (NIT) Warangal** Warangal, India
Bachelor of Technology in Electrical and Electronics Engineering. May 2024

ACHIEVEMENTS

- Secured 99.3rd percentile in JEE Mains (India) - Top 0.7% among 1 million+ candidates
- Recognised with Top 100 Projects at NASA AMES Space Settlement Contest, San Juan, Puerto Rico (2017)
- Solved 300+ Data Structures and Algorithms problems (Medium-Hard difficulty) on LeetCode